Calibration Is Not Style Robustness: Primal–Dual Counterfactual RLCD with Conformal Selective Risk Control

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Abstract

Automation-facing decision models expose typed probabilities, such as $P(\text{yes})$, to accept, defer, or route candidate answers. While Reinforcement Learning for Calibrated Decisions (RLCD) optimizes strictly proper scoring rules to align reported probabilities with empirical accuracy, we prove and show empirically that aggregate calibration does not imply style robustness: a proper-score predictor over observed features can achieve low mixture calibration error while remaining systematically overconfident on stylistically decorated wrong responses. Adapting RM-Bench into a decision-native protocol (*Style-CalibBench*), we separate Extreme-Corner Accuracy (ECA)—concise-correct vs. markdown-incorrect—from official RM-Bench Hard. On Qwen2.5-1.5B, single-head RLCD reaches mixture ECE 0.089 yet degrades to markdown ECE 0.171 and ECA 0.536, while temperature scaling cuts SFT's ECE (0.168→0.083) without moving ECA (0.560). An identity audit further confirms that a detached two-stage style head shifts content logits by $\leq 2.6 \times 10^{-6}$ (0 ECA flips), proving detached auxiliary heads cannot induce content robustness. To achieve causal debiasing on the Pareto frontier, we propose *Counterfactual Constrained RLCD* (**CC-RLCD**), a primal–dual multi-objective framework minimizing within-label style dispersion and conflict margins under an ε-bounded Brier budget, paired with prompt-clustered Conformal Risk Control (CRC). On 1.5B, CC-RLCD reaches ECA 0.694 (+15.8 points over RLCD) and markdown ECE 0.098 while preserving content accuracy (0.742), dominating scalarization, PCGrad, GRL, and CDA, and restoring $\leq 5\%$ selective risk on markdown; across Qwen2.5 (0.5B/1.5B/7B) and Llama-3.1-8B it adds +14.1 to +16.6 ECA points.

1 Introduction

Agentic and automation pipelines increasingly replace free-form LLM judges with typed decision interfaces that emit calibrated probabilities—such as $P(\text{yes})$ for substantive validity—in one forward pass (TypeSafe AI 2026; Guo et al. 2026; Gao et al. 2026). Because downstream code branches directly on p_c to auto-accept, abstain, or escalate, probability calibration is an operational requirement (Guo et al. 2017; Epstein et al. 2026).

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Calibration on an observational test mixture is nevertheless an incomplete reliability criterion. Preference and reward models routinely conflate verbosity, markdown structure, sycophancy, and jargon with substance (Liu et al. 2025; Bharadwaj et al. 2026; Kim, Oh, and Lee 2026). Training a decision readout with a strictly proper score (Brier 1950; Gneiting and Raftery 2007), as in RLCD, elicits the observational conditional $P(y_c{=}1 \mid x, r)$, which absorbs any stylistic cue $s(r)$ correlated with training labels. When style and correctness conflict at deployment, an aggregate-calibrated gate silently accepts formatted errors.

Resolving this conflict requires framing content calibration and counterfactual style invariance as a multi-objective optimization (MOO) problem (Sener and Koltun 2018; Gu et al. 2026; Xu et al. 2026), where optimization mechanics dictate whether debiasing is causal. First, monotone post-hoc scaling fixes mixture ECE without altering cross-style ordering. Second, detaching an auxiliary style head via a stop-gradient mathematically freezes the content pathway, yielding zero causal gain over its Stage-1 checkpoint. Third, unconstrained scalarization or Counterfactual Data Augmentation (CDA) buys cross-style accuracy by sacrificing calibration and overall accuracy (Das and Dennis 1997).

We propose **CC-RLCD** (*Counterfactual Constrained RLCD*; Figure 1a), which solves a constrained Lagrangian primal–dual program (Chamon and Ribeiro 2020) that updates the shared trunk with counterfactual invariance and conflict-margin losses subject to an ε-bounded content Brier budget, followed by a detached Stage-2 style diagnostic head and prompt-clustered Conformal Risk Control (CRC) (Angelopoulos et al. 2024). Our contributions are:

- **Theory & Protocol.** We formalize *mixture-calibration cancellation* (Proposition 1) and introduce *Style-CalibBench*, separating ECA from official RM-Bench Easy/Normal/Hard and mixture from slice ECE.
- **Causal Identity Audit.** Using a matched Stage-1 control, we verify that detached style-head training alters content logits by $\leq 2.6 \times 10^{-6}$ (0 ECA flips), preventing misattribution of initialization variance.
- **Primal–Dual CC-RLCD & CRC Gate.** We couple counterfactual probability invariance and conflict margins under an adaptive dual proper-score budget, paired with prompt-clustered CRC for finite-sample selective

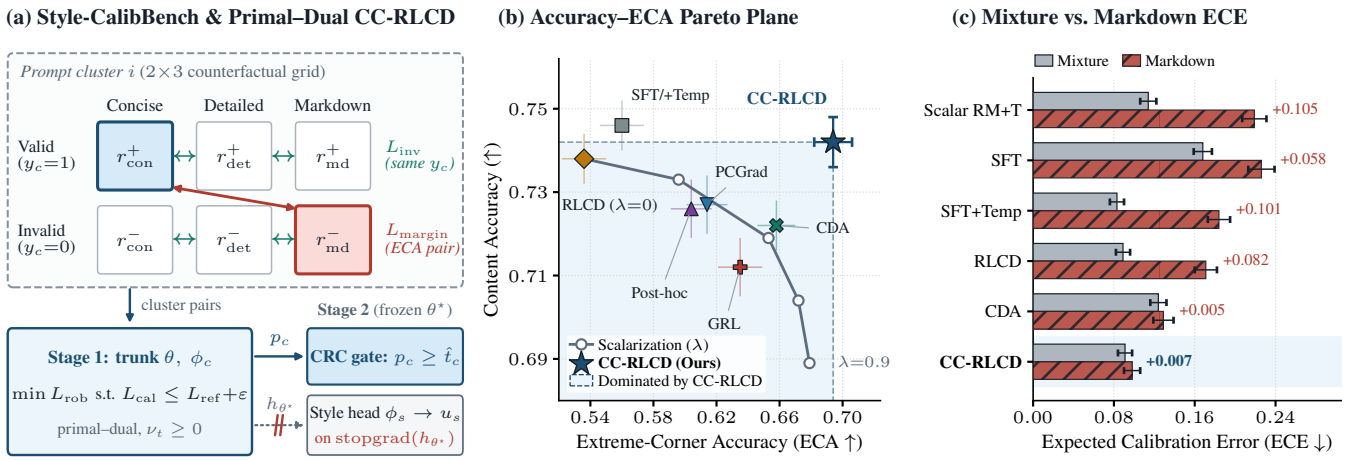


Figure 1: **Framework and 1.5B Main Results.** (a) *Style-CalibBench* crosses content validity and style (2×3); Stage 1 of **CC-RLCD** updates (θ, ϕ_c) via primal–dual counterfactual invariance (L_{inv}) and conflict-margin (L_{margin}) losses under an ε-bounded Brier budget, while Stage 2 fits a detached style head ϕ_s and CRC threshold $\hat{t}_c$. (b) Accuracy–ECA Pareto plane (error crosses: 1 SE over 3 seeds; the scalarization path starts at RLCD, λ=0): every linear scalarization and MOO baseline falls in the shaded region Pareto-dominated by CC-RLCD. (c) Mixture ECE vs. markdown slice ECE (annotated with $+\Delta$ = MD-ECE − ECE): SFT+Temp and RLCD suffer severe slice inflation, whereas CC-RLCD achieves low error on both (+0.007).

risk guarantees.

- **Multi-Scale & OOD Evidence.** Across four backbones (0.5B–8B), ten baselines, and unseen *Flattery* shifts, CC-RLCD reaches the Pareto knee (Figure 1b) and eliminates markdown selective-risk violations.

2 Related Work

Style Bias in Preference Models. RM-Bench reveals sharp reward-model drops when concise correct answers face markdown errors (Liu et al. 2025), and counterfactual audits expose length, jargon, and sycophancy biases (Bharadwaj et al. 2026). Mitigations span causal augmentation (Kim, Oh, and Lee 2026), post-hoc bias regression (Huang et al. 2025), arena rescaling (Zhu et al. 2025), susceptibility reweighting (Wang, Donoway, and Murfet 2026), and RLHF calibration (Leng et al. 2025). Unlike scalar rankers, typed decision models must preserve calibrated probabilities $p_c \in [0, 1]$ for selective gating.

Calibrated Decisions, MOO, and Risk Control. Strictly proper scores elicit true conditionals (Brier 1950; Gneiting and Raftery 2007) for System-One decision models (Type-Safe AI 2026; Guo et al. 2026; Gao et al. 2026; Deußer et al. 2026), yet remain vulnerable to shift-induced overconfidence (Epstein et al. 2026). Multi-objective alignment uses gradient surgery (Yu et al. 2020), adversarial reversal (Ganin and Lempitsky 2015), value decorrelation (Xu et al. 2026), or Pareto filtering (Gu et al. 2026). To avoid non-convex scalarization trade-offs (Das and Dennis 1997), we pair primal–dual constrained learning (Chamon and Ribeiro 2020) with Conformal Risk Control (Angelopoulos et al. 2024).

3 Theory and Style-CalibBench Protocol

3.1 Why Proper Scoring Fails Under Style Shift

Let prompt X, response R, substantive validity $Y_c \in \{0, 1\}$, and style $S \in \{\text{con}, \text{det}, \text{md}\}$ follow a training mixture where formatting correlates with quality. Minimizing expected Brier or log loss drives $p_c(X, R)$ to $P_{\text{train}}(Y_c{=}1 \mid X, S)$ rather than interventional validity $P(Y_c{=}1 \mid \text{do}(S{=}s), X)$.

Proposition 1 (Mixture-Calibration Cancellation)**.** *Let test style $S \in \{\text{con}, \text{md}\}$ be uniform and independent of content truth $P_{\text{test}}(Y_c{=}1 \mid X) = \mu(X)$ with $\mu(X) \sim \text{Unif}[\delta, 1-\delta]$ for $\delta \in (0, \frac{1}{4})$. A style-skewed predictor $p_c(X, \text{md}) = \mu(X) + \delta$ and $p_c(X, \text{con}) = \mu(X) - \delta$ satisfies $\text{ECE}_{\text{mix}} = 2\delta^2/(1-2\delta) < \text{ECE}_{\text{md}} = \delta$ via cross-slice bin cancellation on $[2\delta, 1-2\delta]$, inflates wrong-answer confidence by $+2\delta$, and inverts every conflict pair with $0 < \mu(x^+) - \mu(x^-) < 2\delta$. Moreover, any strictly increasing map $g : [0, 1] \to [0, 1]$ leaves ECA invariant.*

Proposition 1 explains why (i) pooled reliability diagrams conceal severe markdown overconfidence, and (ii) temperature scaling repairs mixture ECE while leaving ECA unchanged because $g(p_c(x, r^+_{\text{con}})) > g(p_c(x, r^-_{\text{md}})) \iff p_c(x, r^+_{\text{con}}) > p_c(x, r^-_{\text{md}})$.

3.2 Style-CalibBench and Metric Separation

Style-CalibBench adapts all 1,327 RM-Bench prompts (Liu et al. 2025) into 6 typed instances per prompt ($y_c \in \{1, 0\} \times s \in \{\text{con}, \text{det}, \text{md}\}$). On safety prompts, y_c=1 marks appropriate refusal or safe compliance. Prompt-level hashing (seed 20261109) yields 929 train, 133 calibration, and 265 test prompts (1,590 test instances) across Chat (90/13/26), Code (160/23/45), Math (370/53/106), Safety-refuse (199/28/57), and Safety-response (110/16/31). Dual-annotator review of

240 rewrites (κ=0.76) flagged 19 concise items dropping minor math qualifiers; we check stability with and without them.

We evaluate four distinct axes: (1) **Calibration**: 15-bin mixture ECE and markdown slice ECE (MD-ECE); (2) **Cross-style ranking**: *Extreme-Corner Accuracy* ($\mathrm{ECA} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}[p_c(x_i, r^{+}_{i,\mathrm{con}}) > p_c(x_i, r^{-}_{i,\mathrm{md}})]$) alongside official RM-Bench *Hard* (averaging $\mathrm{con}^{+} \succ \mathrm{det}^{-}$, $\mathrm{det}^{+} \succ \mathrm{md}^{-}$, and ECA); (3) **Stability**: within-(x_i, y_c) 0.5-threshold *Flip* rate and wrong-answer *Confidence Inflation* $\mathbb{E}[p_c \mid 0, \mathrm{md}] - \mathbb{E}[p_c \mid 0, \mathrm{con}]$; (4) **Selective risk**: test error and coverage under CRC threshold $\hat{t}_c$, with 2,000-resample prompt-bootstrap 95% CIs.

4 Counterfactual Constrained RLCD

4.1 Causal Boundary of Detached Heads

Let $h_\theta(x, r)$ denote the shared trunk representation, $u_c = \mathrm{softmax}(\phi_c(h_\theta)) \in \Delta_2$ the binary content distribution ($p_c = u_{c,1}$), and $u_s = \mathrm{softmax}(\phi_s(h_\theta)) \in \Delta_5$ a 1–5 match score for a declared target style s^*, whose gold level is fixed by the (response, target) style pair: 5 if equal, 3 for det/md, 2 for con/det, 1 for con/md. A detached two-stage recipe minimizes Brier risk $L_{\mathrm{cal}}(\theta, \phi_c) = \mathbb{E}[\|u_c - e_{y_c}\|_2^2]$ in Stage A and $L_s(\mathrm{stopgrad}(h_{\theta_A}), \phi_s)$ in Stage B. Because $\nabla_{\theta, \phi_c} L_s \equiv 0$, Stage B enforces $(\theta_B, \phi_{c,B}) = (\theta_A, \phi_{c,A})$ and $p_c^B(x, r) = p_c^A(x, r)$ up to floating-point noise. Moving p_c onto the Pareto knee requires counterfactual gradients in θ subject to explicit calibration protection.

4.2 Stage 1: Primal–Dual Counterfactual Optimization

Within each training prompt cluster i, let $z_c = \log \frac{p_c}{1-p_c}$ denote the content log-odds and $\mathcal{P}_{\mathrm{adv}} = \{(\mathrm{con}, \mathrm{det}), (\mathrm{det}, \mathrm{md}), (\mathrm{con}, \mathrm{md})\}$ the style-conflict pairs (m=1.0). We define within-label invariance and cross-style margin losses:

$$L_{\mathrm{inv}} = \mathbb{E}_{i, y_c, s \neq s'}\big[(p_c(x_i, r^{y_c}_{i,s}) - p_c(x_i, r^{y_c}_{i,s'}))^2\big], \quad (1)$$

$$L_{\mathrm{margin}} = \mathbb{E}_{i,(a,b)}\big[\max\big(0,\, m - \Delta z_c(x_i; a, b)\big)\big],$$

where $\Delta z_c(x_i; a, b) = z_c(x_i, r^{+}_{i,a}) - z_c(x_i, r^{-}_{i,b})$ for $(a, b) \in \mathcal{P}_{\mathrm{adv}}$. Setting $L_{\mathrm{rob}} = L_{\mathrm{inv}} + \beta L_{\mathrm{margin}}$ and anchoring to a Stage-0 RLCD Brier risk L_{ref}, we solve the constrained program:

$$\min_{\theta, \phi_c} L_{\mathrm{rob}}(\theta, \phi_c) \quad \text{s.t.} \quad L_{\mathrm{cal}}(\theta, \phi_c) \leq L_{\mathrm{ref}} + \varepsilon, \quad (2)$$

via primal–dual updates on $\mathcal{L}(\theta, \phi_c, \nu) = L_{\mathrm{rob}}(\theta, \phi_c) + \nu(L_{\mathrm{cal}}(\theta, \phi_c) - L_{\mathrm{ref}} - \varepsilon)$ with projected dual ascent:

$$\nu_{t+1} = \big[\nu_t + \eta_\nu\big(\widehat{L}_{\mathrm{cal}}(\theta_t, \phi_{c,t}) - L_{\mathrm{ref}} - \varepsilon\big)\big]_+. \quad (3)$$

Unlike fixed scalarization $(1-\lambda)L_{\mathrm{cal}} + \lambda L_{\mathrm{rob}}$, ν_t spikes only when debiasing threatens Brier honesty and relaxes inside $L_{\mathrm{ref}} + \varepsilon$, recovering non-convex Pareto knees (Chamon and Ribeiro 2020).

Method (1.5B)	Acc↑	Hard↑	ECA↑	ECE↓	MD-ECE↓	Flip↓
Readout	0.547	0.407	0.358	0.241	0.327	0.519
Scalar RM+T	0.618	0.459	0.412	0.114	0.219	0.463
SFT	**0.746**	0.611	0.560	0.168	0.226	0.337
SFT+Temp	**0.746**	0.611	0.560	**0.083**	0.184	0.337
Post-hoc Reg.	0.726	0.648	0.604	0.097	0.142	0.241
RLCD (Brier)	0.738	0.576	0.536	0.089	0.171	0.281
Detached 2-H*	0.738	0.576	0.536	0.089	0.171	0.281
PCGrad	0.727	0.637	0.614	0.111	0.156	0.214
GRL (Adv.)	0.712	0.662	0.635	0.119	0.138	0.189
CDA	0.722	0.684	0.658	0.124	0.129	0.171
CC-RLCD	0.742	**0.716**	**0.694**	0.091	**0.098**	**0.138**
w/o L_{inv}	0.736	0.695	0.671	0.098	0.115	0.169
w/o L_{margin}	0.740	0.668	0.639	0.090	0.109	0.147
Fixed λ=0.5	0.719	0.679	0.653	0.116	0.125	0.164
Log-score	0.740	0.711	0.688	0.093	0.102	0.142

Table 1: Test means (3 seeds) on Qwen2.5-1.5B. *Initialized from RLCD: Stage-2 detached style training (style QWK 0.72) leaves p_c identical ($\max|\Delta z_c| = 2.6 \times 10^{-6}$).

4.3 Stage 2 Diagnostic Head & Conformal Gate

Stage 2 fits ϕ_s on $\mathrm{stopgrad}(h_{\theta^\star}(x, r))$ via 5-level ordinal Brier loss, reporting u_s as an audit signal while preserving $p_c^\star$. For deployment gating on M=133 calibration clusters, let $\hat{R}_i(t) \in [0, 1]$ be the selective error on accepted items in cluster i, monotonized as $\sup_{t' \geq t}$ so that it is non-increasing in t. Prompt-clustered Conformal Risk Control (Angelopoulos et al. 2024) sets:

$$\hat{t}_c = \inf\Big\{t \in [0,1] : \frac{1}{M+1}\Big(\sum_{i=1}^{M} \hat{R}_i(t) + 1\Big) \leq \alpha\Big\}, \quad (4)$$

ensuring $\mathbb{E}[R_{\mathrm{test}}(\hat{t}_c)] \leq \alpha$ over exchangeable prompt clusters.

5 Experiments

Setup and Baselines. We train Qwen2.5-1.5B-Instruct (LoRA r=16, lr 2×10^{-5}, length 1024, ε=0.005, β=0.5, η_ν=0.05, 2+1 epochs, seeds {17, 23, 41}) and replicate on Qwen2.5-0.5B, Qwen2.5-7B, and Llama-3.1-8B-Instruct. Baselines include: **Readout**, **Scalar RM+T** (Skywork-Reward-Llama-3.1-8B + temperature), **SFT**, **SFT+Temp**, **Post-hoc Reg.** (local length/markdown bias regression (Huang et al. 2025)), **RLCD** (Brier), **Detached 2-Head** (matched Stage-1 identity control), **PCGrad** (Yu et al. 2020), **GRL** (Ganin and Lempitsky 2015), and **CDA** (Kim, Oh, and Lee 2026).

5.1 Identity Audit and Main Comparison

Identity Audit. Initialized from the exact Stage-1 RLCD checkpoint across all 1,590 test items, Detached 2-Head reaches style quadratic-weighted κ (QWK) 0.72 against the gold match levels while changing content logits by at most 2.6×10^{-6} (mean 3.4×10^{-7}, 0 ECA flips; Table 1), proving that unmatched two-head gains reflect Stage-1 variance rather than Stage-2 supervision.

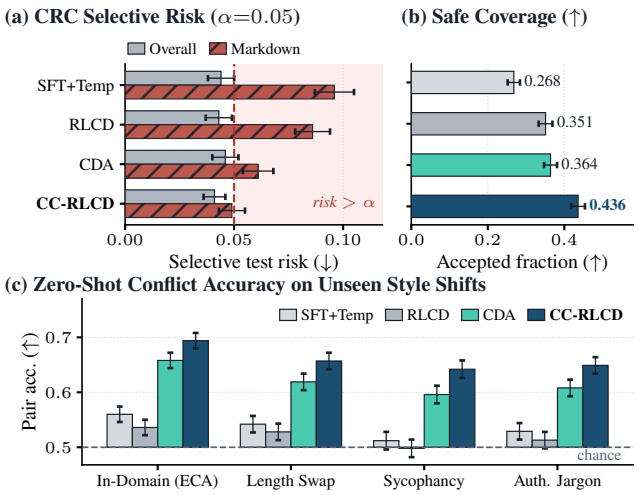


Figure 2: **Selective Risk Control and Unseen Style Shifts (1.5B).** (a–b) Prompt-clustered CRC (α=0.05; all bars: 1 SE over 3 seeds) bounds overall risk across methods, but only CC-RLCD keeps markdown risk $\leq 5\%$ (0.049) while maximizing safe coverage (0.436). (c) Zero-shot conflict-pair accuracy on three unseen *Flattery* perturbations; bars start at chance (0.5).

Pareto Dominance over Baselines. Table 1 and Figure 1b–c corroborate Proposition 1. RLCD achieves mixture ECE 0.089 [0.067, 0.116] yet its markdown ECE is $1.9\times$ higher (0.171; concise slice 0.061), with ECA 0.536 [0.479, 0.592]. SFT+Temp hits mixture ECE 0.083 without moving ECA (0.560). Post-hoc regression lifts ECA to 0.604, while shared-trunk baselines (PCGrad, GRL, CDA) reach at most ECA 0.658 (CDA) only by lowering accuracy (0.712–0.727) and raising ECE (0.111–0.124). **CC-RLCD** achieves ECA 0.694 [0.642, 0.746] (+0.158 over RLCD, $p<0.001$) and official Hard 0.716, cuts markdown ECE to 0.098 and flip rate to 0.138, and preserves accuracy (0.742) and mixture ECE (0.091). Wrong-answer confidence inflation drops from +0.083 to +0.019 ($\rho(p_c, \mathbb{I}_{\text{md}} \mid y_c{=}0)$: 0.23→0.04). Ablations confirm L_{inv} drives flip stability, L_{margin} drives ECA (+5.5 pts), and adaptive dual $\nu^\star \approx 1.84$ beats fixed scalarization. Excluding the 19 drifted concise items changes every method's ECA by at most 0.012 and preserves the ranking.

5.2 Conformal Risk, OOD Shifts, and Scaling

Conformal Selective Prediction. Naive calibration thresholding on 133 prompts overshoots α=0.05 on test data (RLCD risk 0.061, SFT 0.078). Prompt-clustered CRC (Eq. 4; Figure 2a–b) bounds overall risk (≤ 0.046) for all runs, yet RLCD and SFT+Temp still breach 5% on markdown (0.086, 0.096) due to formatted overconfidence. CC-RLCD satisfies $\leq 5\%$ both overall (0.041) and on markdown (0.049) while lifting safe coverage from 0.351 to 0.436. Adding style-head gating cuts coverage (0.362) without risk reduction.

Slices, Unseen Styles, and Scale. CC-RLCD improves 1.5B ECA in every domain: Chat (0.547→0.688), Code (0.608→0.765), Math (0.490→0.651), Safety-refuse (0.541→0.723), and Safety-response (0.571→0.690). On three unseen OOD perturbations from Bharadwaj et al. (2026)—*Length Swap*, *Sycophancy*, and *Authoritative Jargon* (Figure 2c)—RLCD hovers near chance (0.498–0.528) whereas CC-RLCD retains 0.642–0.657 (+12.9 to +14.4 pts). Table 2 confirms that 7B/8B scale does not eliminate style bias under RLCD (MD-ECE 0.152–0.158), while CC-RLCD adds +14.1 to +16.6 ECA points on every backbone.

Backbone	Method	Acc↑	ECA↑	ECE↓	MD-ECE↓
Qwen2.5-0.5B	RLCD	0.688	0.493	0.117	0.204
Qwen2.5-0.5B	CC-RLCD	**0.701**	**0.634**	**0.114**	**0.126**
Qwen2.5-7B	RLCD	0.784	0.581	**0.074**	0.152
Qwen2.5-7B	CDA	0.769	0.708	0.106	0.114
Qwen2.5-7B	CC-RLCD	**0.789**	**0.747**	0.077	**0.083**
Llama-3.1-8B	RLCD	0.779	0.574	**0.079**	0.158
Llama-3.1-8B	CC-RLCD	**0.783**	**0.739**	0.081	**0.086**

Table 2: Multi-scale and cross-family evaluation (3 seeds). Scaling improves accuracy yet RLCD's markdown ECE stays 1.7–2.1$\times$ its mixture ECE, which CC-RLCD resolves.

6 Limitations and Conclusion

Style-CalibBench uses structured rewrites rather than organic traffic, and 265 test prompts yield $\pm$5–6-pt bootstrap half-widths on ECA; formal Mondrian group-conditional CRC requires larger per-style splits, and Stage-1 requires open weights. Overall, mixture calibration does not imply style robustness, and neither post-hoc scaling nor detached heads alter cross-style rankings. Coupling primal–dual counterfactual training with CRC yields calibrated, style-robust decision probabilities.

Ethical and AI Use Statement

We release split hashes, primal–dual training code, and CRC scripts under upstream RM-Bench terms; auxiliary style scores are audit diagnostics, not truth labels. LLMs assisted with prose editing and plotting.

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